Discrete Flow Maps

Abstract. The sequential nature of autoregressive next-token prediction imposes a fundamental speed limit on Large Language Models. While continuous flow models offer a path to parallel generation, they traditionally demand expensive iterative integration. Flow Maps bypass this bottleneck by compressing generative trajectories into single-step mappings—theoretically enabling the generation of full text sequences from noise in a single forward pass. However, standard formulations rely on Euclidean regression losses that are geometrically ill-suited for discrete data. In this work, we resolve this conflict with Discrete Flow Maps, a framework that reconciles trajectory compression with the geometry of the probability simplex. We recast standard flow map training for the discrete domain, aligning the training dynamics with the discrete nature of language. Empirically, this strict geometric alignment allows our method to surpass previous state-of-the-art results in discrete flow modeling.

1 Introduction

Large Language Models (LLMs) have fundamentally altered the computational landscape. Yet, despite their staggering utility, the field remains constrained by a structural bottleneck: the inherently sequential nature of next-token prediction. The dominant architecture, autoregressive (AR) modeling, generates text one step at a time. While this approach has scaled remarkably, it imposes a linear computational cost on generation, rendering long-form reasoning and real-time synthesis prohibitively expensive.

The community has responded with various optimization techniques—such as speculative decoding and multi-token prediction—aimed at extracting efficiency from the AR backbone. However, these methods remain strictly bound by the underlying serial nature of AR models. To fundamentally overcome this limitation, we must look beyond the next-token prediction paradigm entirely.

Separately, diffusion models and flow matching have matured into the dominant paradigms for generative synthesis in continuous domains. By modeling generation as the transformation of noise into data via differential equations, these frameworks offer a rigorous path toward non-autoregressive, parallel generation. Crucially, they unlock capabilities that AR models lack, such as precise test-time steering, iterative refinement, and flexible guidance mechanisms. To accelerate these models, recent techniques such as consistency models and flow maps have emerged, learning to map any point on a generative trajectory directly to its endpoint, thereby compressing the iterative integration process into a single forward pass. The recent surge in such distillation methods suggests a tantalizing possibility: training flow models on text to achieve massive speedups while retaining powerful control mechanisms—potentially revolutionizing our approach to complex reasoning tasks.

However, a fundamental geometric mismatch stands in the way. Standard flow map objectives are natively designed for Euclidean space $\mathbb{R}^K$, relying heavily on L^2 regression losses. Text, in contrast, is discrete. Prior research has consistently demonstrated that training generative models for discrete data with regression losses yields suboptimal performance compared to likelihood-based alternatives such as cross-entropy. Simply put, treating a probability distribution like a coordinate in Euclidean space is a fundamental misalignment; the geometry of the loss does not match the geometry of the data.

In this work, we resolve this conflict by recasting flow map training to fully leverage the discrete structure of the target distribution. We propose a framework where parameterizations and objective functions are geometrically consistent with the probability simplex, systematically translating the governing dynamics of continuous flow maps to the discrete domain. We replace standard regression with cross-entropy and KL divergence objectives, thereby aligning the training dynamics with the discrete nature of language.

While recent works, such as Categorical Flow Maps, have aimed to adapt flow maps to discrete data, they often rely on composite loss bounds or inexact objectives to approximate consistency. By contrast, we provide exact training objectives derived directly from the geometric definitions of the flow. We validate this theoretical rigor empirically,

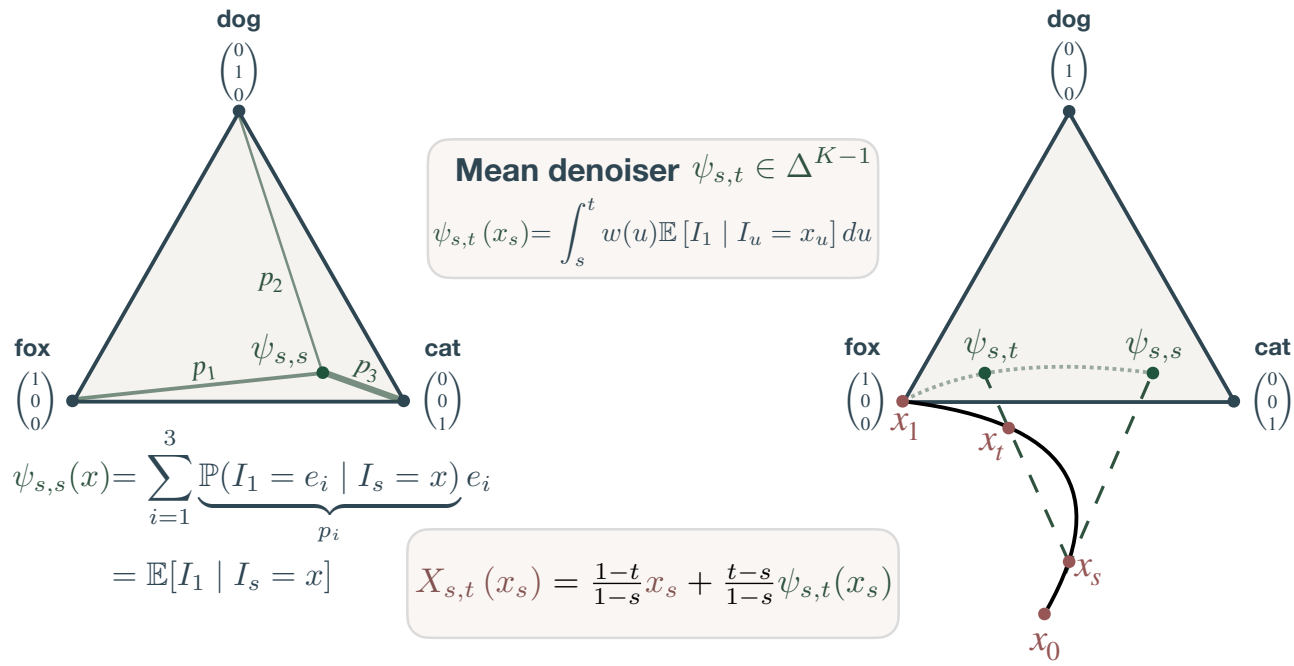

Figure 1 ***Overview of the geometry of the discrete flow map. Left:*** *The instantaneous denoiser $\psi_{s,s}(x_s)$ at any time s is always on the simplex by weighted sum.* ***Right:*** *By derivation of equation 19, the* ***mean denoiser*** *$\psi_{s,t}(x_s)$ is also always a projection onto the simplex, making possible a direct parameterization of the flow map $X_{s,t}$ that enables our cross-entropy loss functions.*

demonstrating that respecting the geometry of the simplex allows us to surpass previous state-of-the-art results in discrete flow modeling.

2 Preliminaries

We begin by formalizing language modeling within the geometry of the probability simplex, alongside the standard autoregressive formulation. We then review continuous generative flows that transport probability mass from noise to data and admit efficient acceleration via flow maps that compress trajectories into single-step operators.

2.1 Language Modeling on the Simplex

Let $\mathcal{V} = \{e_1, \ldots, e_K\} \subset \mathbb{R}^K$ be the set of standard basis vectors representing a finite vocabulary of size K. These vectors constitute the vertices of the $(K-1)$-dimensional probability simplex $\Delta^{K-1} := \{x \in \mathbb{R}^K : x \geq 0, \langle \mathbf{1}, x \rangle = 1\}$. Consequently, a sequence of discrete tokens of length L can be represented as a matrix $\mathbf{x} = (x^1, \ldots, x^L) \in \mathcal{V}^L$. The objective of language modeling is to learn a probability distribution supported on these discrete sequences $\mathcal{V}^L$ that approximates the true data distribution, which we denote by $p_1(\mathbf{x})$.

Autoregressive Modeling. The standard Autoregressive (AR) approach factorizes the joint distribution $p_1(\mathbf{x})$ into a product of conditional probabilities via the chain rule:

$$p_1(\mathbf{x}) = \prod_{\ell=1}^{L} p(x^\ell \mid x^{<\ell}), \tag{1}$$

where $x^{<\ell}$ denotes the tokens preceding position ℓ. AR models parametrize these conditionals using a neural network that outputs a categorical distribution over $\mathcal{V}$ at each step. To learn these conditional probabilities, training minimizes the negative log-likelihood, or equivalently, the cross-entropy loss.

Cross Entropy. This objective is fundamental to discrete generative modeling as it allows for learning a probability mass function over a finite vocabulary. Let Y be a discrete random variable taking values in $\mathcal{V}$

(representing one-hot encoded tokens), and let X be a conditioning random variable taking values in a generic measurable space $\mathcal{X}$. We optimize over the space of measurable functions $f : \mathcal{X} \to \Delta^{K-1}$ to find the minimizer of the expected cross-entropy loss:

$$\mathcal{L}(f) = \mathbb{E}_{X,Y}\left[-\sum_{k=1}^{K} Y^{(k)} \log f^{(k)}(X)\right]. \tag{2}$$

The global minimizer f^* of this objective is the *conditional expectation* of the target:

$$f^*(x) = \mathbb{E}[Y \mid X = x] \tag{3}$$

Crucially, since Y is a one-hot vector, its expectation corresponds exactly to the vector of class probabilities: $\mathbb{E}[Y^{(k)} \mid X = x] = \mathbb{P}(Y = e_k \mid X = x)$. Thus, minimizing cross-entropy directly recovers the true conditional probability mass function. In the context of AR modeling, identifying X with the history $x^{<\ell}$ and Y with the next token x^ℓ, this loss forces the model to learn $p(x^\ell \mid x^{<\ell})$, the true conditional distribution of the next token given the previous ones. While cross entropy provides a rigorous foundation for training discrete generative models, the standard AR framework is hampered by the sequential dependency in equation 1, necessitating slow serial generation. To overcome this computational bottleneck, we instead turn to continuous generative flows, which we formalize next.

2.2 Continuous Generative Flows

We instead approach generative modeling by defining a neural transport map from a simple noise distribution p_0 to the complex data distribution p_1. This transport is realized via an Ordinary Differential Equation (ODE) defined by a velocity field (drift) $b_t : \mathbb{R}^K \to \mathbb{R}^K$:

$$\dot{x}_t = b_t(x_t), \qquad x_0 \sim p_0, \tag{4}$$

constructed such that the trajectory endpoints x_1 follow p_1. To learn this transport, we first specify the desired evolution of marginal densities using a *stochastic interpolant*. We define a process I_t that linearly interpolates between a noise sample $I_0 \sim p_0$ and a data sample $I_1 \sim p_1$:

$$I_t = (1-t)I_0 + tI_1. \tag{5}$$

This process implicitly defines a time-dependent density $p_t := \mathrm{Law}(I_t)$ connecting p_0 to p_1. We seek a vector field b_t such that the marginal path of the ODE equation 4 matches the interpolant (i.e., $x_t \sim p_t$). The optimal choice for b_t is the conditional expectation of the interpolant's velocity:

$$b_t(x) = \mathbb{E}[I_1 - I_0 \mid I_t = x]. \tag{6}$$

To learn this drift, we parametrize a neural network $\hat{b}_t : \mathbb{R}^K \to \mathbb{R}^K$ and minimize the flow matching objective:

$$b = \operatorname*{argmin}_{\hat{b}} \int_0^1 \mathbb{E}\left[\left\|\hat{b}_t(I_t) - (I_1 - I_0)\right\|^2\right] dt, \tag{7}$$

where the expectation is taken over the interpolant process.

2.3 Flow Maps and Trajectory Compression

Simulating equation 4 during inference requires numerical integration, necessitating numerous evaluations of the neural drift b_t. To circumvent this bottleneck, methods such as Consistency Models and Flow Maps compress these continuous trajectories into single-step and few-step mappings.

Formally, a flow map $X_{s,t} : \mathbb{R}^K \to \mathbb{R}^K$ acts as the solution operator for the probability flow ODE. For any valid trajectory $(x_t)_{t\in[0,1]}$, the map satisfies:

$$X_{s,t}(x_s) = x_t, \quad \forall s, t \in [0,1]. \tag{8}$$

In essence, $X_{s,t}$ jumps directly between times s and t along the flow. For training, it is standard to parametrize the flow map in residual form via the average velocity $v_{s,t}(x)$:

$$X_{s,t}(x) = x + (t-s)\, v_{s,t}(x). \tag{9}$$

For the flow map to remain consistent with the underlying ODE dynamics, the average velocity must converge to the instantaneous drift as the time step vanishes. This is formalized as the *tangent condition*:

$$\lim_{s \to t} \partial_t X_{s,t}(x) = v_{t,t}(x) = b_t(x). \tag{10}$$

We enforce this condition by training a neural parameterization $\hat{v}_{s,t}$ to match the interpolant's velocity along the diagonal $s = t$, yielding the standard diagonal loss:

$$\mathcal{L}_{\text{diag}}(\hat{v}) = \int_0^1 \mathbb{E} \left\| \hat{v}_{t,t}(I_t) - (I_1 - I_0) \right\|^2 dt. \tag{11}$$

While the diagonal objective anchors the model $\hat{v}_{t,t}$ to the instantaneous drift b_t, it does not constrain the trajectory for distinct times $s \neq t$. To ensure the learned map forms a valid global trajectory, we must enforce *consistency constraints*. These can be expressed through three equivalent identities:

$$\text{Semi-Group:} \quad X_{u,t}\big(X_{s,u}(x)\big) = X_{s,t}(x), \tag{12a}$$

$$\text{Lagrangian:} \quad \partial_t X_{s,t}(x) = v_{t,t}\big(X_{s,t}(x)\big), \tag{12b}$$

$$\text{Eulerian:} \quad \partial_s X_{s,t}(x) + v_{s,s}(x) \cdot \nabla X_{s,t}(x) = 0, \tag{12c}$$

for all $s, u, t \in [0, 1]$. The Semi-Group rule enforces compositionality, ensuring that the direct transport from s to t is equivalent to the sequential transport through any intermediate time u. The Lagrangian rule dictates that the flow endpoint moves according to the instantaneous drift, while the Eulerian rule ensures invariance to the source time.

To train the model, we employ consistency objectives that directly penalize violations of these identities. We formulate these losses as the squared residuals of the rules in equation 12:

$$\mathcal{L}_{\text{PSD}}(\hat{v}) = \iiint\limits_{0 \le s \le u \le t \le 1} \left\| \hat{X}_{u,t}\big(\hat{X}_{s,u}(x)\big) - \hat{X}_{s,t}(x) \right\|^2 ds du dt, \tag{13}$$

$$\mathcal{L}_{\text{LSD}}(\hat{v}) = \iint\limits_{0 \le s \le t \le 1} \left\| \partial_t \hat{X}_{s,t}(x) - \hat{v}_{t,t}\big(\hat{X}_{s,t}(x)\big) \right\|^2 ds dt, \tag{14}$$

$$\mathcal{L}_{\text{ESD}}(\hat{v}) = \iint\limits_{0 \le s \le t \le 1} \left\| \partial_s \hat{X}_{s,t}(x) + \hat{v}_{s,s}(x) \cdot \nabla \hat{X}_{s,t}(x) \right\|^2 ds dt. \tag{15}$$

Our total loss is the sum of the diagonal loss and any of these consistency losses:

$$\mathcal{L}_{\text{total}}(\hat{v}) = \mathcal{L}_{\text{diag}}(\hat{v}) + \mathcal{L}_{\text{cons}}(\hat{v}). \tag{16}$$

We train the neural parametrized average velocity $\hat{v}$ by minimizing $\mathcal{L}_{\text{total}}$, with the minimizer yielding $\hat{v}_{s,t} = v_{s,t}$, and so $\hat{X}_{s,t}$ recovers the true flow map $X_{s,t}$ at optimality.

3 Discrete Flow Maps

We now adapt the flow map framework to the discrete domain. For clarity, we formulate our method for distributions p_1 supported on the vocabulary $\mathcal{V} \subset \mathbb{R}^K$. The extension to sequences of length L (i.e., distributions on $\mathcal{V}^L$) is immediate by applying these operations position-wise. Standard flow map objectives force discrete data into a Euclidean regression framework, minimizing L^2 errors that are geometrically ill-suited for probability distributions. In this work, we resolve this misalignment by grounding the entire flow map framework within the geometry of the probability simplex Δ^{K-1}. We adopt a parametrization that naturally respects the simplex and consistency objectives based on cross-entropy and KL divergence.

3.1 The Mean Denoiser Parametrization

Standard flow maps parametrize the trajectory $X_{s,t}$ via the unconstrained average velocity $v_{s,t} : \mathbb{R}^K \to \mathbb{R}^K$. While effective in Euclidean space, this formulation ignores the geometry of discrete data: even if the target

distribution p_1 is supported on the simplex, the velocity $v_{s,t}$ need not, and effectively can be any vector in $\mathbb{R}^K$. We seek instead to reparametrize the flow map in terms of an object that explicitly resides on the simplex. We achieve this by defining the flow via the *mean denoiser* $\psi_{s,t} : \mathbb{R}^K \to \Delta^{K-1}$, related to the average velocity by:

$$v_{s,t}(x) = \frac{\psi_{s,t}(x) - x}{1 - s}. \tag{17}$$

Substituting this expression into the flow map update $X_{s,t}(x) = x + (t - s)v_{s,t}(x)$ yields the convex combination:

$$X_{s,t}(x) = \frac{1-t}{1-s}x + \frac{t-s}{1-s}\psi_{s,t}(x). \tag{18}$$

Crucially, the restriction of $\psi_{s,t}$ to the probability simplex is not merely a heuristic architectural choice; rather, it is a geometric necessity of the optimal flow.

Mean Denoiser. The mean denoiser $\psi_{s,t}(x)$ is the time-averaged conditional expectation of the data:

$$\psi_{s,t}(x_s) = \int_s^t w(u)\mathbb{E}[I_1 \mid I_u = x_u]\, du, \tag{19}$$

where $(x_\tau)_{\tau\in[s,t]}$ is a trajectory of the flow and $w(u) = \frac{(1-s)(1-t)}{(t-s)(1-u)^2}$ is a probability density on $[s, t]$.

Since the conditional expectation $\mathbb{E}[I_1 \mid I_u]$ is an expectation of one-hot vectors, it always lies on the simplex. Consequently, $\psi_{s,t}$—as a convex combination of these expectations—must also strictly reside on the simplex. Architecturally, we enforce this by parametrizing a neural network with unconstrained logits $\hat{z}_{s,t} : \mathbb{R}^K \to \mathbb{R}^K$ and defining $\hat{\psi}_{s,t}(x) = \mathrm{Softmax}(\hat{z}_{s,t}(x))$. This ensures the predicted target is always a valid distribution $\hat{\psi} \in \Delta^{K-1}$.

Extension to General Polytopes. Although we focus on the probability simplex, this framework generalizes to any convex polytope $\mathcal{P}$ defined by linear constraints [**PeterH:** I would say the general principle is: If the data distribution p_1 lies in $\Lambda \subset \mathbb{R}^K$, then the mean denoiser $\psi_{s,t} \in \mathrm{ConvexHull}(\Lambda)$. In our case, $\Lambda = \{e_1, \cdots, e_K\}$ and $\mathrm{ConvexHull}(\Lambda) = \Delta^{K-1}$]. As shown in Proposition A, if the data support lies within $\mathcal{P}$, then the range of the mean denoiser $\psi_{s,t}$ remains within $\mathcal{P}$ as well. Architecturally, one can replace the Softmax with a suitable link function to ensure $\hat{\psi}_{s,t} \in \mathcal{P}$ by design.

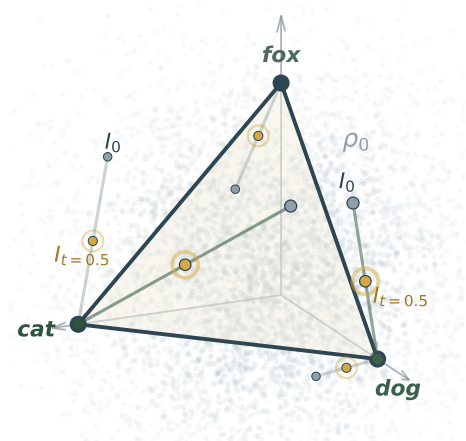

Figure 2 *Interpolants with initial point $I_0 \in \mathbb{R}^d$ from ρ_0 randomly connecting to simplex vertices.*

3.2 Training Objectives

We now turn to training $\hat{\psi}_{s,t}$. Since $\hat{\psi}_{s,t}$ is constrained to output valid probability distributions, the usual flow map training objectives—originally formulated as Euclidean regression losses—can be reformulated to act on distributions. This allows us to use cross-entropy and KL-divergence losses that respect the geometry of the simplex. As in standard flow map training, we employ two complementary objectives: a diagonal loss that anchors $\hat{\psi}_{t,t}$, and a consistency loss that enforces geometric validity for $s < t$.

3.2.1 Diagonal Loss

We begin with the following identity (see **?**) which states that the diagonal of the mean denoiser is just the standard denoiser:

Diagonal Identity for $\psi_{s,t}$. For any $t \in [0, 1]$, the simplex predictor satisfies:

$$\psi_{t,t}(x) = \mathbb{E}[I_1 \mid I_t = x]. \tag{20}$$

Based on this identity, we train $\hat{\psi}_{t,t}$ to predict the target class I_1 given the noisy state I_t. Since I_1 takes values in $\mathcal{V}$, we can minimize the expected cross-entropy loss:

$$\mathcal{L}_{\text{diag}}(\hat{\psi}) = \int_0^1 \mathbb{E}\left[-\sum_{k=1}^{K} I_1^{(k)} \log \hat{\psi}_{t,t}^{(k)}(I_t)\right] dt, \tag{21}$$

where the expectation is over the joint distribution of I_t and I_1. Any proper scoring rule or Bregman divergence could alternatively be used here. Minimizing this loss ensures that $\hat{\psi}_{t,t} = \psi_{t,t}$ at optimality.

3.2.2 Consistency Loss

To learn a valid flow map $X_{s,t}$, the model must satisfy the consistency constraints in equation 12. We first rewrite these fundamental flow identities in terms of the mean denoisier $\psi_{s,t}$.

Discrete Flow Map Identities. For any times $s < u < t \in [0, 1]$, each the following identities uniquely characterize $\psi_{s,t}$ in conjunction with equation (18):

Semi-Group: $$\psi_{s,t}(x) = \alpha_{s,u,t}\psi_{s,u}(x) + \beta_{s,u,t}\psi_{u,t}(X_{s,u}(x)), \tag{22}$$

Lagrangian: $$\psi_{s,t}(x) = \psi_{t,t}(X_{s,t}(x)) - \gamma_{s,t}\frac{\partial}{\partial t}\psi_{s,t}(x), \tag{23}$$

Eulerian: $$\frac{\partial}{\partial s}\psi_{s,t}(x) + J_x\psi_{s,t}(x)b_s(x) = \kappa_{s,t}(\psi_{s,t}(x) - \psi_{s,s}(x)), \tag{24}$$

Diff. Semi-Group: $$\frac{d}{du}\left[\alpha_{s,u,t}\psi_{s,u}(x) + \beta_{s,u,t}\psi_{u,t}(X_{s,u}(x))\right] = 0. \tag{25}$$

Here, J_x is the Jacobian with respect to x, the coefficients $\alpha_{s,u,t} = \frac{(u-s)(1-t)}{(t-s)(1-u)}$ and $\beta_{s,u,t} = \frac{(t-u)(1-s)}{(t-s)(1-u)}$ are non-negative and sum to 1, while we define $\gamma_{s,t} = \frac{(t-s)(1-t)}{1-s}$ and $\kappa_{s,t} = \frac{1-t}{(1-s)(t-s)}$.

We now convert these geometric identities into tractable training objectives. Our goal is to leverage the discrete geometry of the model to formulate exact losses.

Consistency via Semi-Group Loss (PSD). We would like to enforce the Semi-Group identity directly on our network $\hat{\psi}$:

$$\hat{\psi}_{s,t}(x) = \alpha_{s,u,t}\hat{\psi}_{s,u}(x) + \beta_{s,u,t}\hat{\psi}_{u,t}(\hat{X}_{s,u}(x)). \tag{26}$$

Since the network output is constrained to the simplex, the right-hand side—a convex combination of probability vectors—defines a valid target distribution. Treating this composite prediction as the teacher, we distill it into the student $\hat{\psi}_{s,t}$ by minimizing the KL divergence:

$$\mathcal{L}_{\text{PSD}}(\hat{\psi}) = \iiint_{0\le s\le u\le t\le 1} \mathbb{E}\left[D_{\text{KL}}\left(\text{sg}\left[\alpha_{s,u,t}\hat{\psi}_{s,u}(I_s) + \beta_{s,u,t}\hat{\psi}_{u,t}(\hat{X}_{s,u}(I_s))\right] \parallel \hat{\psi}_{s,t}(I_s)\right)\right] dsdudt, \tag{27}$$

where sg$[\cdot]$ denotes the stop-gradient operator and where the expectation is over $I_s \sim p_s$. Minimizing this loss enforces equation 29, which, together with the diagonal condition, ensures that $\hat{X}$ recovers the true target flow map.

Next, we discuss how the three flow map consistency rules, the Semi-Group, Lagrangian and Eulerian rules, can be adapted to the discrete setting.

Consistency via Semi-Group Loss. Standard Progressive Distillation enforces the semi-group property $X_{s,t} = X_{u,t} \circ X_{s,u}$. To enable cross-entropy training, we translate this property into a condition on the simplex predictor $\psi_{s,t}$.

Semi-Group Identity for $\psi_{s,t}$. For any $s < u < t \in [0, 1]$, the simplex predictor satisfies:

$$\psi_{s,t}(x_s) = \alpha_{s,u,t}\psi_{s,u}(x_s) + \beta_{s,u,t}\psi_{u,t}(x_u), \tag{28}$$

where $\alpha_{s,u,t} = \frac{(u-s)(1-t)}{(t-s)(1-u)}$ and $\beta_{s,u,t} = \frac{(t-u)(1-s)}{(t-s)(1-u)}$ are non-negative coefficients summing to 1.

We would like to enforce this identity directly on our network $\hat{\psi}$:

$$\hat{\psi}_{s,t}(x_s) = \alpha_{s,u,t}\hat{\psi}_{s,u}(x_s) + \beta_{s,u,t}\hat{\psi}_{u,t}(\hat{X}_{s,u}(x_s)). \tag{29}$$

Since the network output is constrained to the simplex, the right-hand side—a convex combination of probability vectors—defines a valid target distribution. Treating this composite prediction as the teacher, we distill it into the student $\hat{\psi}_{s,t}$ by minimizing the KL divergence:

$$\mathcal{L}_{\text{PSD}}(\hat{\psi}) = \iiint\limits_{0\le s\le u\le t\le 1} \mathbb{E}\left[D_{\text{KL}}\left(\text{sg}\left[\alpha_{s,u,t}\hat{\psi}_{s,u}(I_s) + \beta_{s,u,t}\hat{\psi}_{u,t}(\hat{X}_{s,u}(I_s))\right] \parallel \hat{\psi}_{s,t}(I_s)\right)\right] dsdudt, \tag{30}$$

where sg[·] denotes the stop-gradient operator and where the expectation is over $I_s \sim p_s$. Minimizing this loss enforces equation 29, which, together with the diagonal condition, ensures that $\hat{X}$ recovers the true target flow map.

Consistency via Lagrangian Loss. Alternatively, consistency can be enforced via the Lagrangian perspective, which requires the flow endpoint velocity to match the instantaneous drift: $\partial_t X_{s,t}(x_s) = v_{t,t}(X_{s,t}(x_s))$. We express this constraint in terms of the simplex predictor $\psi_{s,t}$.

Lagrangian Identity for $\psi_{s,t}$. For any $s < t \in [0, 1]$, the simplex predictor satisfies the ODE:

$$\psi_{s,t}(x_s) = \psi_{t,t}(x_t) - C_{s,t}\frac{\partial}{\partial t}\psi_{s,t}(x_s), \qquad C_{s,t} = \tfrac{(t-s)(1-t)}{1-s}. \tag{31}$$

For the optimal simplex predictor $\psi_{s,t}$ both sides of the above equation are probability distributions. However, when we attempt to enforce this constraint on our neural network $\hat{\psi}_{s,t}$, the right side might not be a valid probability distribution and so we cannot use distributional losses like the KL divergence between the two sides. To rigorously enforce the Lagrangian identity, we instead derive an equivalent condition in *logit space* and then recast it to probability distributions.

Lagrangian Logit Consistency. Let $\psi_{s,t}(x) = \text{Softmax}(z_{s,t}(x))$. The Lagrangian Identity is equivalent to:

$$\psi_{s,t}(x_s) = \text{Softmax}\left(z_{t,t}(x_t) - \log\left(\mathbf{1} + C_{s,t}(\dot{z}_{s,t}(x_s) - \bar{\dot{z}}_{s,t}(x_s)\mathbf{1})\right)\right), \tag{32}$$

where $\dot{z}_{s,t} = \partial_t z_{s,t}$ and $\bar{\dot{z}}_{s,t} = \langle \psi_{s,t}, \dot{z}_{s,t}\rangle$ is the expected logit velocity.

This identity defines a geometrically valid target for training. We minimize the KL divergence from the student $\hat{\psi}_{s,t}$ to this target:

$$\mathcal{L}_{\text{LSD}}(\hat{\psi}) = \iint\limits_{0\le s\le t\le 1} \mathbb{E}\left[D_{\text{KL}}\left(\text{Softmax}\Big(\text{sg}\left[\hat{z}_{t,t}(\hat{X}_{s,t}(I_s)) - \log(\mathbf{1} + C_{s,t}(\dot{\hat{z}}_{s,t}(I_s) - \bar{\dot{\hat{z}}}_{s,t}(I_s)\mathbf{1}))\right]\Big) \parallel \hat{\psi}_{s,t}(I_s)\right)\right] dsdt, \tag{33}$$

where the expectation is taken over $I_s \sim p_s$. Minimizing this loss ensures that both Lagrangian consistency rules are satisfied by $\hat{\psi}_{s,t}$.

Consistency via Eulerian Loss. Finally, we consider the Eulerian perspective, $\partial_s X_{s,t}(x_s) = 0$, which enforces invariance with respect to the source time s, and we again translate this rule for the simplex predictor.

Eulerian Identity for $\psi_{s,t}$. For any $s < t \in [0,1]$, the simplex predictor satisfies the PDE:

$$\frac{\partial}{\partial s}\psi_{s,t}(x) + J_x\psi_{s,t}(x)b_s(x) = \frac{1-t}{(1-s)(t-s)}(\psi_{s,t}(x) - \psi_{s,s}(x)), \tag{34}$$

where $J_x\psi_{s,t}(x)$ is the Jacobian with respect to x.

As with the Lagrangian case, we can derive an equivalent condition in logit space.

Eulerian Logit Consistency. Let $\psi_{s,t}(x) = \mathrm{Softmax}(z_{s,t}(x))$. The Eulerian Identity is equivalent to:

$$\psi_{s,t}(x) = \mathrm{Softmax}\left(z_{s,s}(x) - \log\left(\mathbf{1} - \kappa_{s,t}^{-1}(D_s z_{s,t}(x) - \overline{D}_s z_{s,t}(x)\mathbf{1})\right)\right), \tag{35}$$

where $\kappa_{s,t} = \frac{1-t}{(1-s)(t-s)}$, $D_s z_{s,t} = \partial_s z_{s,t} + J_x z_{s,t} b_s$ is the total derivative of the logits along the drift b_s, and $\overline{D}_s z_{s,t} = \langle \psi_{s,t}, D_s z_{s,t}\rangle$ is the expected total derivative.

Differentiate the PSD condition with respect to u to get

$$\frac{d}{du}X_{u,t}(X_{s,u}(x)) = 0. \tag{36}$$

Expanding in terms of the average velocity we have

$$\frac{d}{du}\left(x + (u-s)v_{s,u}(x_s) + (t-u)v_{u,t}(X_{s,u}(x))\right) = 0, \tag{37}$$

which yields

$$(u-s)\partial_u v_{s,u}(x_s) + v_{s,u}(x_s) + (t-u)Jv_{u,t}(X_{s,u}(x))\partial_u X_{s,u}(x) - v_{u,t}(X_{s,u}(x)) = 0. \tag{38}$$

Setting $u = s$ recovers ESD and setting $u = t$ recovers LSD.

4 Tilt Matching

To generate from a reward-tilted distribution $p_1^*(x) \propto p_1(x)e^{r(x)}$, the optimal instantaneous predictor must match the conditional Esscher transform:

$$\psi_{t,t}^*(x) = \frac{\mathbb{E}[e^{r(I_1)}I_1 \mid I_t = x]}{\mathbb{E}[e^{r(I_1)} \mid I_t = x]}. \tag{39}$$

While the reward-weighted cross-entropy loss, $\mathbb{E}[e^{r(I_1)}\mathrm{CE}(I_1 \parallel \hat{\psi}_{t,t}(I_t))]$, admits $\psi_{t,t}^*$ as its unique global minimizer, we will construct a lower variance objective by using a control variate. Let $\psi_{t,t}(I_t) = \mathbb{E}[I_1 \mid I_t]$ denote the prediction of a frozen base model. We define the target as a convex combination governed by $c \in [0,1]$:

$$T_c(I_t, I_1) = \frac{c}{e^{r(I_1)}}\psi_{t,t}(I_t) + \left(1 - \frac{c}{e^{r(I_1)}}\right)I_1. \tag{40}$$

If $r \geq 0$, then $T_c \in \Delta^{K-1}$. Therefore we will use the objective

$$\mathcal{L}_{\mathrm{Tilt}}(\hat{\psi}) = \int_0^1 \mathbb{E}\left[e^{r(I_1)}\mathrm{CE}(T_c(I_t, I_1) \parallel \hat{\psi}_{t,t}(I_t))\right]dt, \tag{41}$$

whose minimizer recovers $\psi_{t,t}^*$.

5 Experiments

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A Proofs

Diagonal Identity for $\pi_{s,t}$. For any $t \in [0,1]$, the simplex predictor satisfies:

$$\psi_{t,t}(x) = \mathbb{E}[I_1 \mid I_t = x]. \tag{42}$$

Proof. The flow map $X_{s,t}$ is defined by $X_{s,t}(x) = x + (t-s)v_{s,t}(x)$, with $v_{s,t}(x) = \frac{\pi_{s,t}(x)-x}{1-s}$. Taking the limit $s \to t$, the instantaneous velocity field b_t generating the trajectory satisfies:

$$b_t(x) = \lim_{s\to t} \frac{X_{s,t}(x) - x}{t-s} = \lim_{s\to t} \frac{\pi_{s,t}(x) - x}{1-s} = \frac{\pi_{t,t}(x) - x}{1-t}. \tag{43}$$

Using the relations $b_t(x) = \mathbb{E}[I_1 - I_0 \mid I_t = x]$ and $x = (1-t)\,\mathbb{E}[I_0 \mid I_t = x] + t\,\mathbb{E}[I_1 \mid I_t = x]$, gives

$$b_t(x) = \frac{\mathbb{E}[I_1 \mid I_t = x] - x}{1-t}. \tag{44}$$

Equating equation 43 and equation 44 completes the proof. □

Linear Constraint Preservation. For $c \in \mathbb{R}, a \in \mathbb{R}^d$, let the data support $\mathrm{supp}(p_1)$ be bounded by a linear inequality $a^\top I_1 \geq c$, such that $I_1 \sim p_1$ satisfies this constraint almost surely. Then the partial denoiser also satisfies $a^\top \pi_{s,t}(x) \geq c$ for all $s, t \in [0,1]$.

Proof. Fix x_0 and consider the trajectory $(x_t)_{t\in[0,1]}$ arising from the drift b_t. Let $y(t) = \langle a, x_t\rangle$ be the projection of the flow trajectory onto the vector a. Using equation 44, the evolution of $y(t)$ is governed by:

$$\dot{y}(t) = \frac{m(t) - y(t)}{1-t}, \tag{45}$$

where $m(t) = \mathbb{E}[\langle a, I_1\rangle \mid I_t = x]$. Since $\langle a, I_1\rangle \geq c$ almost surely, we have $m(t) \geq c$. The partial denoiser is defined in terms of the flow map as:

$$\pi_{s,t}(x_s) = x_s + (1-s)v_{s,t}(x_s) = x_s + \frac{1-s}{t-s}(x_t - x_s). \tag{46}$$

Projecting this onto a gives the scalar relation:

$$\langle a, \pi_{s,t}(x_s)\rangle = y(s) + \frac{1-s}{t-s}(y(t) - y(s)). \tag{47}$$

To solve for $y(t)$, we use equation 45 to express the ODE as:

$$\frac{d}{dt}\left(\frac{y(t)}{1-t}\right) = \frac{(1-t)\dot{y}(t) + y(t)}{(1-t)^2} = \frac{m(t)}{(1-t)^2}. \tag{48}$$

Integrating from s to t yields:

$$\frac{y(t)}{1-t} - \frac{y(s)}{1-s} = \int_s^t \frac{m(u)}{(1-u)^2} du. \tag{49}$$

Rearranging for $y(t)$ and substituting into equation 47 leads to exact cancellation of the $y(s)$ terms:

$$\langle a, \pi_{s,t}(x_s) \rangle = \frac{(1-s)(1-t)}{t-s} \int_s^t \frac{m(u)}{(1-u)^2} du. \tag{50}$$

We now bound the integral using the property $m(u) \geq c$:

$$\int_s^t \frac{m(u)}{(1-u)^2} du \geq c \int_s^t \frac{du}{(1-u)^2} = c \left[\frac{1}{1-u} \right]_s^t = c \left(\frac{1}{1-t} - \frac{1}{1-s} \right) = c \frac{t-s}{(1-s)(1-t)}. \tag{51}$$

Substituting this back into the expression for the projected denoiser:

$$\langle a, \pi_{s,t}(x_s) \rangle \geq \left[\frac{(1-s)(1-t)}{t-s} \right] \cdot \left[c \frac{t-s}{(1-s)(1-t)} \right] = c. \tag{52}$$

Thus, $\langle a, \pi_{s,t}(x_s) \rangle \geq a$. □

Simplex Preservation. Let the data support lie on the probability simplex $\Delta^{K-1} = \{x \in \mathbb{R}^K : x \geq 0, \langle \mathbf{1}, x \rangle = 1\}$. If $I_1 \in \Delta^{K-1}$ almost surely, then the partial denoiser satisfies $\pi_{s,t}(x) \in \Delta^{K-1}$ for all $s, t \in [0, 1]$.

Proof. The probability simplex is defined as $\Delta^{K-1} = \{x \in \mathbb{R}^K \mid x \geq 0, \langle \mathbf{1}, x \rangle = 1\}$. We verify that $\pi_{s,t}$ satisfies these conditions by invoking the Linear Constraint Preservation. First, for non-negativity, we choose $a = e_k$, the one-hot vector with a 1 in the kth position, and $c = 0$. Since the data satisfies $\langle e_k, I_1 \rangle \geq 0$ almost surely, the proposition implies $\pi_{s,t}^{(k)}(x) = \langle e_k, \pi_{s,t}(x) \rangle \geq 0$ for all $k = 1, \ldots, K$. Second, for normalization, we apply the proposition twice. With $a = \mathbf{1}$ and $c = 1$, we obtain $\langle \mathbf{1}, \pi_{s,t}(x) \rangle \geq 1$. Conversely, with $a = -\mathbf{1}$ and $c = -1$ (since $\langle -\mathbf{1}, I_1 \rangle = -1 \geq -1$), we obtain $\langle -\mathbf{1}, \pi_{s,t}(x) \rangle \geq -1$, which implies $\langle \mathbf{1}, \pi_{s,t}(x) \rangle \leq 1$. Together, these inequalities force $\langle \mathbf{1}, \pi_{s,t}(x) \rangle = 1$. Satisfying both conditions implies $\pi_{s,t}(x) \in \Delta^{K-1}$. □

Semi-Group Identity for $\pi_{s,t}$. For any $s < u < t \in [0, 1]$, the simplex predictor satisfies:

$$\psi_{s,t}(x_s) = \alpha_{s,u,t} \psi_{s,u}(x_s) + \beta_{s,u,t} \psi_{u,t}(x_u), \tag{53}$$

where $\alpha_{s,u,t} = \frac{(u-s)(1-t)}{(t-s)(1-u)}$ and $\beta_{s,u,t} = \frac{(t-u)(1-s)}{(t-s)(1-u)}$ are non-negative coefficients summing to 1.

Proof. We rewrite the flow map $X_{s,t}(x) = x + (t-s)\frac{\pi_{s,t}(x) - x}{1-s}$ as:

$$X_{s,t}(x) = \frac{1-t}{1-s} x + \frac{t-s}{1-s} \pi_{s,t}(x). \tag{54}$$

By the semi-group property $X_{s,t} = X_{u,t} \circ X_{s,u}$ for $s < u < t$, we expand the composition using equation 54:

$$\begin{aligned}
\frac{1-t}{1-s} x + \frac{t-s}{1-s} \pi_{s,t}(x) = X_{s,t}(x) &= X_{u,t}(X_{s,u}(x)) && (55)\\
&= \frac{1-t}{1-u} X_{s,u}(x) + \frac{t-u}{1-u} \pi_{u,t}(X_{s,u}(x)) && (56)\\
&= \frac{1-t}{1-u} \left[\frac{1-u}{1-s} x + \frac{u-s}{1-s} \pi_{s,u}(x) \right] + \frac{t-u}{1-u} \pi_{u,t}(X_{s,u}(x)) && (57)\\
&= \frac{1-t}{1-s} x + \frac{(1-t)(u-s)}{(1-u)(1-s)} \pi_{s,u}(x) + \frac{t-u}{1-u} \pi_{u,t}(X_{s,u}(x)). && (58)
\end{aligned}$$

Solving for $\pi_{s,t}(x)$ yields:

$$\pi_{s,t}(x) = \underbrace{\frac{(1-t)(u-s)}{(1-u)(t-s)}}_{\alpha_{s,u,t}} \pi_{s,u}(x) + \underbrace{\frac{(1-s)(t-u)}{(1-u)(t-s)}}_{\beta_{s,u,t}} \pi_{u,t}(X_{s,u}(x)). \tag{59}$$

A simple check shows that $\alpha_{s,u,t} + \beta_{s,u,t} = 1$. □

Lagrangian Identity for $\pi_{s,t}$. For any $s < t \in [0,1]$, the simplex predictor satisfies the ODE:

$$\psi_{s,t}(x_s) = \psi_{t,t}(x_t) - C_{s,t}\frac{\partial}{\partial t}\psi_{s,t}(x_s), \qquad C_{s,t} = \tfrac{(t-s)(1-t)}{1-s}. \tag{60}$$

Proof. We differentiate with respect to t the flow map definition $X_{s,t}(x_s) = x_s + (t-s)v_{s,t}(x_s)$, where $v_{s,t}(x_s) = \frac{\pi_{s,t}(x_s)-x_s}{1-s}$:

$$\frac{\partial}{\partial t}X_{s,t}(x_s) = v_{s,t}(x_s) + (t-s)\frac{\partial}{\partial t}v_{s,t}(x_s) = \frac{\pi_{s,t}(x_s)-x_s}{1-s} + \frac{t-s}{1-s}\frac{\partial}{\partial t}\pi_{s,t}(x_s). \tag{61}$$

By definition, the flow map trajectory is generated by the instantaneous drift field $b_t(x_t) = \frac{\pi_{t,t}(x_t)-x_t}{1-t}$:

$$\frac{\partial}{\partial t}X_{s,t}(x_s) = \frac{\partial}{\partial t}x_t = b_t(x_t) = \frac{\pi_{t,t}(x_t)-x_t}{1-t}. \tag{62}$$

Equating equation 61 and equation 62, multiplying by $(1-t)$, and substituting $x_t = x_s + (t-s)\frac{\pi_{s,t}(x_s)-x_s}{1-s}$:

$$\pi_{t,t}(x_t) - \left[x_s + (t-s)\frac{\pi_{s,t}(x_s)-x_s}{1-s}\right] = \frac{1-t}{1-s}(\pi_{s,t}(x_s)-x_s) + \frac{(1-t)(t-s)}{1-s}\frac{\partial}{\partial t}\pi_{s,t}(x_s). \tag{63}$$

Simplifying, we obtain

$$\pi_{t,t}(x_t) = \pi_{s,t}(x_s) + C_{s,t}\frac{\partial}{\partial t}\pi_{s,t}(x_s). \tag{64}$$

Rearranging for $\pi_{s,t}$ completes the proof. □

Lagrangian Logit Consistency. Let $\psi_{s,t}(x) = \mathrm{Softmax}(z_{s,t}(x))$. The Lagrangian Identity is equivalent to:

$$\psi_{s,t}(x_s) = \mathrm{Softmax}\left(z_{t,t}(x_t) - \log\left(\mathbf{1} + C_{s,t}(\dot{z}_{s,t}(x_s) - \bar{\dot{z}}_{s,t}(x_s)\mathbf{1})\right)\right), \tag{65}$$

where $\dot{z}_{s,t} = \partial_t z_{s,t}$ and $\bar{\dot{z}}_{s,t} = \langle \psi_{s,t}, \dot{z}_{s,t}\rangle$ is the expected logit velocity.

Proof. We start from the Lagrangian identity:

$$\pi_{t,t}(x_t) = \pi_{s,t}(x_s) + C_{s,t}\frac{\partial}{\partial t}\pi_{s,t}(x_s). \tag{66}$$

We first derive the time derivative of $\pi_{s,t}(x_s) = \mathrm{Softmax}(z_{s,t}(x_s))$. For the k-th component $\pi_{s,t}^{(k)} = \frac{\exp(z_{s,t}^{(k)})}{\sum_j \exp(z_{s,t}^{(j)})}$, the derivative is:

$$\frac{\partial \pi_{s,t}^{(k)}}{\partial t} = \pi_{s,t}^{(k)}\dot{z}_{s,t}^{(k)} - \pi_{s,t}^{(k)}\sum_j \pi_{s,t}^{(j)}\dot{z}_{s,t}^{(j)} = \pi_{s,t}^{(k)}(\dot{z}_{s,t}^{(k)} - \bar{\dot{z}}_{s,t}), \tag{67}$$

where $\bar{\dot{z}}_{s,t} = \langle \pi_{s,t}, \dot{z}_{s,t}\rangle$. In vector notation, this is $\partial_t \pi_{s,t} = \pi_{s,t} \odot (\dot{z}_{s,t} - \bar{\dot{z}}_{s,t}\mathbf{1})$. Substituting this derivative back into the Lagrangian identity:

$$\pi_{t,t}(x_t) = \pi_{s,t}(x_s) + C_{s,t}\Big[\pi_{s,t}(x_s) \odot (\dot{z}_{s,t}(x_s) - \bar{\dot{z}}_{s,t}(x_s)\mathbf{1})\Big]. \tag{68}$$

Factoring out $\pi_{s,t}(x_s)$ yields:

$$\pi_{t,t}(x_t) = \pi_{s,t}(x_s) \odot \Big(\mathbf{1} + C_{s,t}(\dot{z}_{s,t}(x_s) - \bar{\dot{z}}_{s,t}(x_s)\mathbf{1})\Big). \tag{69}$$

We now take the element-wise logarithm of both sides. We recall that logits are defined only up to an arbitrary additive scalar function, i.e., $\log \pi_{s,t}(x) = z_{s,t}(x) + c_{s,t}(x)\mathbf{1}$ for some scalar function $c_{s,t}(x)$. Therefore, the equality in probability space implies equality in logit space modulo this scalar shift:

$$z_{t,t}(x_t) + c_{t,t}(x_t)\mathbf{1} = z_{s,t}(x_s) + c_{s,t}(x)\mathbf{1} + \log\Big(\mathbf{1} + C_{s,t}(\dot{z}_{s,t}(x_s) - \bar{\dot{z}}_{s,t}(x_s)\mathbf{1})\Big). \tag{70}$$

Rearranging to solve for the student logits $z_{s,t}$ and applying the Softmax function (which is invariant to the scalar shift) proves the equivalence:

$$\pi_{s,t}(x_s) = \text{Softmax}\left(z_{t,t}(x_t) - \log\left(\mathbf{1} + C_{s,t}(\dot{z}_{s,t}(x_s) - \bar{\dot{z}}_{s,t}(x_s)\mathbf{1})\right)\right). \tag{71}$$

□

Eulerian Identity for $\pi_{s,t}$. For any $s < t \in [0,1]$, the simplex predictor satisfies the PDE:

$$\frac{\partial}{\partial s}\psi_{s,t}(x) + J_x\psi_{s,t}(x)b_s(x) = \frac{1-t}{(1-s)(t-s)}(\psi_{s,t}(x) - \psi_{s,s}(x)), \tag{72}$$

where $J_x\psi_{s,t}(x)$ is the Jacobian with respect to x.

Proof. We start with the defining parametrization of the partial denoiser from equation **??**:

$$\frac{1-t}{1-s}x_s + \frac{t-s}{1-s}\pi_{s,t}(x_s) = x_s + (t-s)\frac{\pi_{s,t}(x_s) - x_s}{1-s} = X_{s,t}(x_s) = x_t. \tag{73}$$

We take the derivative with respect to s, noting that $\frac{d}{ds}x_s = b_s(x_s)$, $\frac{d}{ds}(\frac{1-t}{1-s}) = \frac{1-t}{(1-s)^2}$, and $\frac{d}{ds}(\frac{t-s}{1-s}) = \frac{t-1}{(1-s)^2}$:

$$\frac{1-t}{1-s}b_s(x_s) + \frac{1-t}{(1-s)^2}x_s + \frac{t-s}{1-s}\frac{d}{ds}\pi_{s,t}(x_s) + \frac{t-1}{(1-s)^2}\pi_{s,t}(x_s) = 0. \tag{74}$$

We group the terms involving x_s and $\pi_{s,t}(x_s)$ to identify the secant velocity $v_{s,t}(x_s) = \frac{\pi_{s,t}(x_s) - x_s}{1-s}$, and rewrite the equation as:

$$\frac{1-t}{1-s}b_s(x_s) - \frac{1-t}{1-s}\underbrace{\left[\frac{\pi_{s,t}(x_s) - x_s}{1-s}\right]}_{v_{s,t}(x_s)} + \frac{t-s}{1-s}\frac{d}{ds}\pi_{s,t}(x_s) = 0. \tag{75}$$

Rearranging to isolate the total derivative $\frac{d}{ds}\pi_{s,t}$:

$$\frac{t-s}{1-s}\frac{d}{ds}\pi_{s,t}(x_s) = \frac{1-t}{1-s}(v_{s,t}(x_s) - b_s(x_s)). \tag{76}$$

We use the definition of the instantaneous drift $b_s(x_s) = \frac{\pi_{s,s}(x_s) - x_s}{1-s}$ to obtain:

$$v_{s,t}(x_s) - b_s(x_s) = \frac{\pi_{s,t}(x_s) - x_s}{1-s} - \frac{\pi_{s,s}(x_s) - x_s}{1-s} = \frac{\pi_{s,t}(x_s) - \pi_{s,s}(x_s)}{1-s}. \tag{77}$$

Substituting this back and multiplying by $\frac{1-s}{t-s}$:

$$\frac{d}{ds}\pi_{s,t}(x_s) = \frac{1-t}{(t-s)(1-s)}(\pi_{s,t}(x_s) - \pi_{s,s}(x_s)). \tag{78}$$

Expanding the total derivative yields the result. □

Eulerian Logit Consistency. Let $\psi_{s,t}(x) = \text{Softmax}(z_{s,t}(x))$. The Eulerian Identity is equivalent to:

$$\psi_{s,t}(x) = \text{Softmax}\left(z_{s,s}(x) - \log\left(\mathbf{1} - \kappa_{s,t}^{-1}(D_s z_{s,t}(x) - \overline{D}_s z_{s,t}(x)\mathbf{1})\right)\right), \tag{79}$$

where $\kappa_{s,t} = \frac{1-t}{(1-s)(t-s)}$, $D_s z_{s,t} = \partial_s z_{s,t} + J_x z_{s,t} b_s$ is the total derivative of the logits along the drift b_s, and $\overline{D}_s z_{s,t} = \langle \psi_{s,t}, D_s z_{s,t} \rangle$ is the expected total derivative.

Proof. We start from the Eulerian identity:

$$\frac{d}{ds}\pi_{s,t}(x) = \kappa_{s,t}(\pi_{s,t}(x) - \pi_{s,s}(x)), \tag{80}$$

where $\kappa_{s,t} = \frac{1-t}{(1-s)(t-s)}$. We first derive the total derivative of $\pi_{s,t}(x_s) = \text{Softmax}(z_{s,t}(x)_s)$ with respect to s along the flow trajectory. For the k-th component $\pi_{s,t}^{(k)}(x_s) = \frac{\exp(z_{s,t}^{(k)}(x_s))}{\sum_j \exp(z_{s,t}^{(j)}(x_s))}$, the derivative is:

$$\frac{d}{ds}\pi_{s,t}^{(k)}(x_s) = \pi_{s,t}^{(k)}(x_s)\frac{d}{ds}z_{s,t}^{(k)}(x_s) - \pi_{s,t}^{(k)}(x_s)\sum_j \pi_{s,t}^{(j)}(x_s)\frac{d}{ds}z_{s,t}^{(j)}(x_s) \tag{81}$$

$$= \pi_{s,t}^{(k)}(x_s)(\frac{d}{ds}z_{s,t}^{(k)}(x_s) - \overline{D}_s z_{s,t}(x_s)), \tag{82}$$

where $\overline{D}_s z_{s,t}(x_s) = \langle \pi_{s,t}(x_s), \frac{d}{ds}z_{s,t}(x_s) \rangle$ is the expected total derivative. In vector notation, this is $\frac{d}{ds}\pi_{s,t}(x_s) = \pi_{s,t}(x_s) \odot (\frac{d}{ds}z_{s,t}(x_s) - \overline{D}_s z_{s,t}(x_s)\mathbf{1})$. Substituting this derivative back into the Eulerian identity:

$$\kappa_{s,t}(\pi_{s,t}(x_s) - \pi_{s,s}(x_s)) = \pi_{s,t}(x_s) \odot (\frac{d}{ds}z_{s,t}(x_s) - \overline{D}_s z_{s,t}(x_s)\mathbf{1}). \tag{83}$$

Rearranging and factoring out $\pi_{s,t}(x_s)$ yields:

$$\pi_{s,s}(x_s) = \pi_{s,t}(x_s) \odot \Big(\mathbf{1} - \kappa_{s,t}^{-1}(\frac{d}{ds}z_{s,t}(x_s) - \overline{D}_s z_{s,t}(x_s)\mathbf{1})\Big). \tag{84}$$

We now take the element-wise logarithm of both sides. We recall that logits are defined only up to an arbitrary additive scalar function, i.e., $\log \pi_{s,t}(x) = z_{s,t}(x) + c_{s,t}(x)\mathbf{1}$ for some scalar function $c_{s,t}(x)$. Therefore, the equality in probability space implies equality in logit space modulo this scalar shift:

$$z_{s,s}(x_s) + c_{s,s}(x_s)\mathbf{1} = z_{s,t}(x_s) + c_{s,t}(x_s)\mathbf{1} + \log\Big(\mathbf{1} - \kappa_{s,t}^{-1}(\frac{d}{ds}z_{s,t}(x_s) - \overline{D}_s z_{s,t}(x_s)\mathbf{1})\Big). \tag{85}$$

Rearranging and applying the Softmax function (which is invariant to the scalar shift) proves the equivalence:

$$\pi_{s,t}(x_s) = \text{Softmax}\left(z_{s,s}(x_s) - \log\left(\mathbf{1} - \kappa_{s,t}^{-1}(\frac{d}{ds}z_{s,t}(x_s) - \overline{D}_s z_{s,t}(x_s)\mathbf{1})\right)\right). \tag{86}$$

Expanding the total derivative $\frac{d}{ds}z_{s,t}(x_s) = \frac{\partial}{\partial s}z_{s,t}(x_s) + J_x z_{s,t}(x_s) b_s(x_s)$ yields the result for any x:

$$\pi_{s,t}(x) = \text{Softmax}\left(z_{s,s}(x) - \log\left(\mathbf{1} - \kappa_{s,t}^{-1}(D_s z_{s,t}(x) - \overline{D}_s z_{s,t}(x)\mathbf{1})\right)\right), \tag{87}$$

where $D_s = \partial_s + b_s^T \nabla_x$ is the total derivative operator. □

B General Interpolants

While the main body focuses on the linear schedule, all results extend to a broader class of interpolants. Let $\alpha_t, \beta_t : [0,1] \to \mathbb{R}$ be C^1 time schedules with $\alpha_t > 0$ for $t < 1$ and endpoint constraints $\alpha_0 = 1, \beta_0 = 0$ and $\alpha_1 = 0, \beta_1 = 1$, and define the stochastic interpolant

$$I_t = \alpha_t I_0 + \beta_t I_1, \qquad I_0 \sim p_0,\ I_1 \sim p_1. \tag{88}$$

Define the schedule-dependent scalars

$$\ell_t := \partial_t \log \alpha_t = \frac{\dot{\alpha}_t}{\alpha_t}, \qquad \lambda_t := \dot{\beta}_t - \beta_t \frac{\dot{\alpha}_t}{\alpha_t} = \dot{\beta}_t - \beta_t \ell_t. \tag{89}$$

For $s < t$, define

$$\Gamma_{s,t} := \frac{\alpha_t}{\alpha_s}, \qquad \Delta_{s,t} := \beta_t - \Gamma_{s,t}\beta_s. \tag{90}$$

Then $\Gamma_{s,t}\Gamma_{t,u} = \Gamma_{s,u}$ and $\Delta_{s,t} = \Gamma_{u,t}\Delta_{s,u} + \Delta_{u,t}$ for $s < u < t$.

Mean denoiser parametrization. We parametrize the flow map by a simplex-valued predictor $\psi_{s,t} : \mathbb{R}^K \to \Delta^{K-1}$ via

$$X_{s,t}(x) := \Gamma_{s,t}x + \Delta_{s,t}\psi_{s,t}(x). \tag{91}$$

Equivalently, in residual form $X_{s,t}(x) = x + (t-s)v_{s,t}(x)$ we have

$$v_{s,t}(x) = \frac{\Gamma_{s,t} - 1}{t-s}x + \frac{\Delta_{s,t}}{t-s}\psi_{s,t}(x), \qquad \psi_{s,t}(x) = \frac{x + (t-s)v_{s,t}(x) - \Gamma_{s,t}x}{\Delta_{s,t}}. \tag{92}$$

Note that in the main paper we assume $\alpha_t = 1 - t, \beta_t = t$ and then this recovers the parameterization in equation (17).

Interpolant drift. The probability-flow drift satisfies

$$b_t(x) := \mathbb{E}[\dot{I}_t \mid I_t = x] = \ell_t x + \lambda_t \mathbb{E}[I_1 \mid I_t = x]. \tag{93}$$

Proof. Differentiate $I_t = \alpha_t I_0 + \beta_t I_1$ to get $\dot{I}_t = \dot{\alpha}_t I_0 + \dot{\beta}_t I_1$. Rewrite $I_0 = (I_t - \beta_t I_1)/\alpha_t$ and substitute:

$$\dot{I}_t = \frac{\dot{\alpha}_t}{\alpha_t} I_t + \Big(\dot{\beta}_t - \beta_t \frac{\dot{\alpha}_t}{\alpha_t}\Big) I_1 \tag{94}$$

$$= \ell_t I_t + \lambda_t I_1. \tag{95}$$

Taking expectation conditional on $I_t = x$ yields equation 93. □

Diagonal identity for $\psi_{s,t}$. For any $t \in [0,1]$,

$$\psi_{t,t}(x) = \mathbb{E}[I_1 \mid I_t = x]. \tag{96}$$

Proof. From equation 92,

$$\begin{aligned} b_t(x) &= \lim_{s \to t} v_{s,t}(x) = \lim_{s \to t} \Big(\frac{\Gamma_{s,t} - 1}{t-s} x + \frac{\Delta_{s,t}}{t-s} \psi_{s,t}(x) \Big) \\ &= \ell_t x + \lambda_t \psi_{t,t}(x), \end{aligned}$$

using $\lim_{s \to t} \frac{\Gamma_{s,t}-1}{t-s} = \ell_t$ and $\lim_{s \to t} \frac{\Delta_{s,t}}{t-s} = \lambda_t$. Equating with equation 93 gives equation 96. □

Mean denoiser. Let $(x_u)_{u\in[s,t]}$ be a trajectory of $\dot{x}_u = b_u(x_u)$. Then

$$\psi_{s,t}(x_s) = \int_s^t w_{s,t}(u)\,\mathbb{E}[I_1 \mid I_u = x_u]\,du, \qquad w_{s,t}(u) := \frac{\alpha_t}{\Delta_{s,t}}\frac{\lambda_u}{\alpha_u}. \tag{97}$$

Moreover $\int_s^t w_{s,t}(u)\,du = 1$. If $\lambda_u/\alpha_u \geq 0$ on $[s,t]$ (equivalently $u \mapsto \beta_u/\alpha_u$ is nondecreasing), then $w_{s,t}$ is a probability density and $\psi_{s,t}(x_s) \in \Delta^{K-1}$.

Proof. By equation 93, the trajectory obeys

$$\frac{d}{du}x_u = \ell_u x_u + \lambda_u \psi_{u,u}(x_u). \tag{98}$$

Thus by the chain rule

$$\frac{d}{du}\Big(\frac{x_u}{\alpha_u}\Big) = \frac{\ell_u x_u + \lambda_u \psi_{u,u}(x_u)}{\alpha_u} - \frac{\dot{\alpha}_u}{\alpha_u^2}x_u = \frac{\lambda_u}{\alpha_u}\psi_{u,u}(x_u). \tag{99}$$

Integrating from s to t and multiplying with α_t gives

$$x_t = \Gamma_{s,t}x_s + \alpha_t \int_s^t \frac{\lambda_u}{\alpha_u}\psi_{u,u}(x_u)\,du. \tag{100}$$

Next, note $\frac{d}{du}(\beta_u/\alpha_u) = \lambda_u/\alpha_u$, hence

$$\alpha_t \int_s^t \frac{\lambda_u}{\alpha_u}\,du = \alpha_t\Big(\frac{\beta_t}{\alpha_t} - \frac{\beta_s}{\alpha_s}\Big) = \beta_t - \frac{\alpha_t}{\alpha_s}\beta_s = \Delta_{s,t}. \tag{101}$$

Normalize the integral term in equation 100 using equation 101 to obtain equation 97. The normalization $\int_s^t w_{s,t} = 1$ follows from equation 101. □

Semi-group identity for $\psi_{s,t}$. For any $s < u < t$,

$$\psi_{s,t}(x) = \omega_{s,u,t}\psi_{s,u}(x) + (1-\omega_{s,u,t})\psi_{u,t}(X_{s,u}(x)), \qquad \omega_{s,u,t} := \frac{\Gamma_{u,t}\Delta_{s,u}}{\Delta_{s,t}}. \tag{102}$$

Moreover $\omega_{s,u,t} + (1-\omega_{s,u,t}) = 1$, and if $u \mapsto \beta_u/\alpha_u$ is non-decreasing, then $0 \leq \omega_{s,u,t} \leq 1$.

Note that for $\alpha_t = 1-t, \beta_t = t$, we get

$$\begin{aligned}
\omega_{s,u,t} &= \frac{\Gamma_{u,t}\Delta_{s,u}}{\Delta_{s,t}} \\
&= \frac{\frac{1-t}{1-u}\Big(u - \frac{1-u}{1-s}s\Big)}{t - \frac{1-t}{1-s}s} \\
&= \frac{\frac{1-t}{1-u}\cdot\frac{u(1-s)-(1-u)s}{1-s}}{\frac{t(1-s)-(1-t)s}{1-s}} \\
&= \frac{\frac{1-t}{1-u}\cdot\frac{u-s}{1-s}}{\frac{t-s}{1-s}} \\
&= \frac{1-t}{1-u}\cdot\frac{u-s}{t-s}
\end{aligned}$$

recovering the semigroup identity from the main paper.

Proof. By semi-group, $X_{s,t} = X_{u,t} \circ X_{s,u}$. Using equation 91,

$$\Gamma_{s,t}x + \Delta_{s,t}\psi_{s,t}(x) = X_{s,t}(x) = X_{u,t}(X_{s,u}(x)) = \Gamma_{u,t}(\Gamma_{s,u}x + \Delta_{s,u}\psi_{s,u}(x)) + \Delta_{u,t}\psi_{u,t}(X_{s,u}(x)).$$

Since $\Gamma_{u,t}\Gamma_{s,u} = \Gamma_{s,t}$, we can substract $\Gamma_{s,t}$ on both sides and obtain

$$\Delta_{s,t}\psi_{s,t}(x) = \Gamma_{u,t}\Delta_{s,u}\psi_{s,u}(x) + \Delta_{u,t}\psi_{u,t}(X_{s,u}(x)).$$

Divide by $\Delta_{s,t}$ to get equation 102. The $\omega_{s,u,t} + (1 - \omega_{s,u,t}) = 1$ property uses $\Delta_{s,t} = \Gamma_{u,t}\Delta_{s,u} + \Delta_{u,t}$. If $u \mapsto \beta_u/\alpha_u$ is non-decreasing, then

$$\frac{\beta_t}{\alpha_t} - \frac{\beta_s}{\alpha_s} \geq 0 \tag{103}$$

$$\Rightarrow \quad \beta_t - \frac{\beta_s}{\alpha_s}\alpha_t \geq 0 \tag{104}$$

$$\Rightarrow \quad \Delta_{s,t} = \beta_t - \Gamma_{s,t}\beta_s \geq 0 \tag{105}$$

This implies that also $\omega_{s,u,t} \geq 0$ and $1 - \omega_{s,u,t} = \Delta_{u,t} \geq 0$. □

Lagrangian identity for $\psi_{s,t}$. For any $s < t$,

$$\psi_{s,t}(x) = \psi_{t,t}(X_{s,t}(x)) - C_{s,t}\partial_t\psi_{s,t}(x), \qquad C_{s,t} := \frac{\Delta_{s,t}}{\lambda_t}. \tag{106}$$

Proof. Differentiate equation 91 in t:

$$\partial_t X_{s,t}(x) = (\partial_t\Gamma_{s,t})x + (\partial_t\Delta_{s,t})\psi_{s,t}(x) + \Delta_{s,t}\partial_t\psi_{s,t}(x).$$

where we used that $X_{s,t}(x) = \Gamma_{s,t}x + \Delta_{s,t}\psi_{s,t}(x)$. Use $\partial_t\Gamma_{s,t} = \ell_t\Gamma_{s,t}$ and

$$\partial_t\Delta_{s,t} = \dot{\beta}_t - \frac{\dot{\alpha}_t}{\alpha_s}\beta_s = \dot{\beta}_t - \frac{\dot{\alpha}_t}{\alpha_t}\beta_t + \frac{\dot{\alpha}_t}{\alpha_t}\beta_t - \frac{\dot{\alpha}_t}{\alpha_s}\beta_s = \lambda_t + \ell_t\Delta_{s,t}$$

to obtain

$$\partial_t X_{s,t}(x) = \ell_t X_{s,t}(x) + \lambda_t\psi_{s,t}(x) + \Delta_{s,t}\partial_t\psi_{s,t}(x). \tag{107}$$

By tangency, $\partial_t X_{s,t}(x) = b_t(X_{s,t}(x)) = \ell_t X_{s,t}(x) + \lambda_t\psi_{t,t}(X_{s,t}(x))$. Equate with equation 107 and rearrange to get equation 106. □

Eulerian identity for $\psi_{s,t}$. For any $s < t$,

$$\partial_s\psi_{s,t}(x) + J_x\psi_{s,t}(x)b_s(x) = \kappa_{s,t}\big(\psi_{s,t}(x) - \psi_{s,s}(x)\big), \qquad \kappa_{s,t} := \frac{\Gamma_{s,t}\lambda_s}{\Delta_{s,t}}. \tag{108}$$

Proof. Use $\partial_s X_{s,t}(x) + J_x X_{s,t}(x)b_s(x) = 0$ and equation 91. Compute $\partial_s\Gamma_{s,t} = -\ell_s\Gamma_{s,t}$ and $\partial_s\Delta_{s,t} = -\Gamma_{s,t}\lambda_s$. Then

$$0 = -\ell_s\Gamma_{s,t}x - \Gamma_{s,t}\lambda_s\psi_{s,t}(x) + \Delta_{s,t}\partial_s\psi_{s,t}(x) + \Gamma_{s,t}b_s(x) + \Delta_{s,t}J_x\psi_{s,t}(x)b_s(x).$$

Using $b_s(x) = \ell_s x + \lambda_s\psi_{s,s}(x)$ cancels the ℓ_s terms and yields

$$\Delta_{s,t}\big(\partial_s\psi_{s,t}(x) + J_x\psi_{s,t}(x)b_s(x)\big) = \Gamma_{s,t}\lambda_s\big(\psi_{s,t}(x) - \psi_{s,s}(x)\big),$$

which gives equation 108. □

Logit consistency. Let $\psi_{s,t}(x) = \mathrm{Softmax}(z_{s,t}(x))$.

Lagrangian logit consistency (general schedule). The Lagrangian identity equation 106 is equivalent to

$$\psi_{s,t}(x) = \mathrm{Softmax}\Big(z_{t,t}(X_{s,t}(x)) - \log\big(\mathbf{1} + C_{s,t}(\dot{z}_{s,t}(x) - \bar{\dot{z}}_{s,t}(x)\mathbf{1})\big)\Big), \tag{109}$$

where $\dot{z}_{s,t} := \partial_t z_{s,t}$ and $\bar{\dot{z}}_{s,t}(x) := \langle \psi_{s,t}(x), \dot{z}_{s,t}(x)\rangle$.

Proof. Differentiate $\psi = \mathrm{Softmax}(z)$ in t:

$$\partial_t \psi = \psi \odot (\dot{z} - \bar{\dot{z}}\mathbf{1}).$$

Substitute into equation 106 and rearrange elementwise to get

$$\psi_{t,t}(X) = \psi \odot (\mathbf{1} + C(\dot{z} - \bar{\dot{z}}\mathbf{1})).$$

Taking elementwise log and using $\log \psi = z + c\mathbf{1}$ for some scalar c yields equation 109. □

Eulerian logit consistency (general schedule). The Eulerian identity equation 108 is equivalent to

$$\psi_{s,t}(x) = \mathrm{Softmax}\Big(z_{s,s}(x) - \log\big(\mathbf{1} - \kappa_{s,t}^{-1}(D_s z_{s,t}(x) - \overline{D_s z}_{s,t}(x)\mathbf{1})\big)\Big), \tag{110}$$

where $D_s z_{s,t} := \partial_s z_{s,t} + J_x z_{s,t} b_s$ and $\overline{D_s z}_{s,t}(x) := \langle \psi_{s,t}(x), D_s z_{s,t}(x)\rangle$.

Proof. Along b_s, $D_s\psi = \psi \odot (D_s z - \overline{D_s z}\mathbf{1})$ and $D_s\psi = \partial_s\psi + J_x\psi b_s$. Substitute into equation 108 and rearrange:

$$\psi_{s,s} = \psi \odot \big(\mathbf{1} - \kappa^{-1}(D_s z - \overline{D_s z}\mathbf{1})\big).$$

Taking elementwise log and using $\log\psi = z + c\mathbf{1}$ yields equation 110. □

B.1 Training objectives under a general schedule

Diagonal loss. By equation 96, train $\hat{\psi}_{t,t}$ with cross-entropy under the general interpolant:

$$\mathcal{L}_{\mathrm{diag}}(\hat{\psi}) = \int_0^1 \mathbb{E}\Big[-\sum_{k=1}^{K} I_1^{(k)} \log \hat{\psi}_{t,t}^{(k)}(I_t)\Big]dt, \qquad I_t = \alpha_t I_0 + \beta_t I_1. \tag{111}$$

Model flow map. Given $\hat{\psi}$, define

$$\hat{X}_{s,t}(x) := \Gamma_{s,t}x + \Delta_{s,t}\hat{\psi}_{s,t}(x), \qquad \hat{b}_s(x) := \ell_s x + \lambda_s \hat{\psi}_{s,s}(x). \tag{112}$$

Consistency via Semi-Group loss (PSD). Enforce equation 102 by KL distillation with teacher $T_{\mathrm{PSD}} := \omega_{s,u,t}\hat{\psi}_{s,u}(I_s) + (1-\omega_{s,u,t})\hat{\psi}_{u,t}(\hat{X}_{s,u}(I_s))$:

$$\mathcal{L}_{\mathrm{PSD}}(\hat{\psi}) = \iiint_{0\le s\le u\le t\le 1} \mathbb{E}\Big[D_{\mathrm{KL}}\big(\mathrm{sg}[T_{\mathrm{PSD}}] \| \hat{\psi}_{s,t}(I_s)\big)\Big]ds\,du\,dt. \tag{113}$$

Consistency via Lagrangian loss (LSD). Using equation 109, define the teacher

$$T_{\mathrm{LSD}} := \mathrm{Softmax}\Big(\mathrm{sg}\big[\hat{z}_{t,t}(\hat{X}_{s,t}(I_s)) - \log\big(\mathbf{1} + C_{s,t}(\partial_t \hat{z}_{s,t}(I_s) - \overline{\partial_t \hat{z}}_{s,t}(I_s)\mathbf{1})\big)\big]\Big), \tag{114}$$

where $\hat{\psi}_{s,t} = \mathrm{Softmax}(\hat{z}_{s,t})$ and $\overline{\partial_t \hat{z}}_{s,t} := \langle \hat{\psi}_{s,t}, \partial_t \hat{z}_{s,t}\rangle$. Minimize

$$\mathcal{L}_{\mathrm{LSD}}(\hat{\psi}) = \iint_{0\le s\le t\le 1} \mathbb{E}\Big[D_{\mathrm{KL}}\big(T_{\mathrm{LSD}} \| \hat{\psi}_{s,t}(I_s)\big)\Big]ds\,dt. \tag{115}$$

Consistency via Eulerian loss (ESD). Using equation 110, define $D_s\hat{z}_{s,t}(x) := \partial_s \hat{z}_{s,t}(x) + J_x \hat{z}_{s,t}(x)\hat{b}_s(x)$ and $\overline{D_s\hat{z}}_{s,t} := \langle \hat{\psi}_{s,t}, D_s\hat{z}_{s,t}\rangle$. The teacher is

$$T_{\mathrm{ESD}} := \mathrm{Softmax}\Big(\mathrm{sg}\big[\hat{z}_{s,s}(I_s) - \log\big(\mathbf{1} - \kappa_{s,t}^{-1}(D_s\hat{z}_{s,t}(I_s) - \overline{D_s\hat{z}}_{s,t}(I_s)\mathbf{1})\big)\big]\Big), \tag{116}$$

and the loss is

$$\mathcal{L}_{\mathrm{ESD}}(\hat{\psi}) = \iint_{0\le s\le t\le 1} \mathbb{E}\Big[D_{\mathrm{KL}}\big(T_{\mathrm{ESD}} \| \hat{\psi}_{s,t}(I_s)\big)\Big] ds\, dt. \tag{117}$$

Reduction to the linear schedule. For $\alpha_t = 1-t$ and $\beta_t = t$, we have $\Gamma_{s,t} = \frac{1-t}{1-s}$, $\Delta_{s,t} = \frac{t-s}{1-s}$, $\lambda_t = \frac{1}{1-t}$, hence $C_{s,t} = \frac{(t-s)(1-t)}{1-s}$ and $\kappa_{s,t} = \frac{1-t}{(1-s)(t-s)}$, recovering the linear-interpolant coefficients and identities. “

B.2 Classifier-free guidance

[**PeterH:** Incomplete]

$$\begin{aligned} b_t^w(x|c) &= w b_t(x|c) + (1-w) b_t(x) \\ &= w l_t x + w\lambda_t \mathbb{E}[I_1|I_t = x, c] + (1-w) l_t x + (1-w)\lambda_t \mathbb{E}[I_1|I_t = x] \\ &= l_t x + \lambda_t (w\mathbb{E}[I_1|I_t = x, c] + (1-w)\mathbb{E}[I_1|I_t = x]) \end{aligned}$$

Let $X_{s,t}^w$ be the flow map corresponding to the vector field $b_t^w(x|c)$. Then define the mean CFG denoiser $\psi_{s,t}^w$ via:

$$X_{s,t}^w(x) := \Gamma_{s,t} x + \Delta_{s,t}\psi_{s,t}^w(x). \tag{118}$$

Then in the same way as before we can show that

$$\begin{aligned} b_t^w(x) &= \lim_{s\to t} v_{s,t}^w(x) = \lim_{s\to t}\Big(\frac{\Gamma_{s,t}-1}{t-s}x + \frac{\Delta_{s,t}}{t-s}\psi_{s,t}^w(x)\Big) \\ &= \ell_t x + \lambda_t \psi_{t,t}^w(x) \end{aligned}$$

In particular, this implies that:

$$\psi_{t,t}^w(x) = w\mathbb{E}[I_1|I_t = x, c] + (1-w)\mathbb{E}[I_1|I_t = x]$$

By an identitcal proof, we obtain

$$\psi_{s,t}^w(x_s) = \int_s^t w_{s,t}(u)\psi_{u,u}^w(x_u)\, du, \qquad w_{s,t}(u) := \frac{\alpha_t}{\Delta_{s,t}}\frac{\lambda_u}{\alpha_u}. \tag{119}$$

[**PeterH:** Proof that $X_{s,1}^w$ is in the simplex for the endpoint.]